BIKE SHARING DATA ANALYSIS USING POWER BI

Project: Analysing Smart City Bike Sharing Data
Tool Used: Microsoft Power BI

1. Project Objective

The objective of this project is to analyse bike-sharing station data across different cities and understand station distribution, bike availability, and station operating status using Power BI.

2. Data Preparation

The dataset was imported into Power BI and checked using Power Query. A fact table (`fact_ bike stations`) and two-dimension tables (`dim_ City` and `dim_ Status`) were created. One-to-many relationships were established to organize the data model.

3. Dashboard Visualizations

The dashboard includes KPI cards, charts, and interactive slicers to analyse:

- Total number of bike stations.
- Total bike stands.
- Available bikes across cities.
- Open and closed station distribution.
- Comparison of available bikes and available bike stands by city.
- City-wise station and bike availability using slicers.

4. Key Findings

- Total bike station records: **[2,856]**
- City with the highest number of stations: **[Lyon]**
- City with the highest number of available bikes: **[Bruxelles_Capitale]**
- Open and closed station distribution: **[open-2650, closed-206]**

5. Conclusion

Power BI was used to transform the bike-sharing dataset into an interactive dashboard. The dashboard helps users compare bike station distribution, available bikes, and station status across cities. The data model and slicers support organized analysis and interactive filtering.

Limitation: The supplied dataset does not contain revenue, cost, or distance-travelled data. Therefore, these metrics were not analysed.